AE 462
Lab 4
Lab Session: Monday, March 2, 2026
Due: Monday, March 9, 2026, 9am CST

This laboratory assignment will focus on using CFD simulations for the purposes of applied analysis of transonic airfoil aerodynamics. The items required for your assignment submission are given here, alongside helpful instructions for completing this lab. As with previous assignments, all plots and presentations of data should include units, axis titles, and legends where appropriate. Provide your responses in a single PDF file. Upload any scripts, spreadsheets, or other work used during data analysis.

Report Instructions:
Using CFD-generated data from this exercise, provide a response to the following:

1) **Comparison to experimental data:**
   a. Provide a co-plot of the RAE 2822 C_p distribution for α = 2.31 deg, showing your CFD results and the experimental data provided.
   b. For this case we are running a compressible flow with RANS turbulence modeling. Describe what features of the flow and the associated C_p distribution would be missing if this simulation were instead conducted assuming:
      i. Incompressible, viscous flow at M = 0.2 instead of M = 0.729.
      ii. Inviscid (Euler) compressible flow at the original freestream conditions.
2) **Airfoil performance analysis:**
   a. Provide plots of your CFD-generated C_l, C_d, and C_m. Show C_l and C_m (on the y axis) as a function of α (on the x axis) and C_d as a function of C_l.
   b. Calculate the lift-curve slope across the linear angle-of-attack region (α = 0 deg to 4 deg). Compare your calculated value to the theoretical value of $C_{l,\alpha} = 2\pi/\beta$, where $\beta = \sqrt{1 - M_\infty^2}$, with $C_{l,\alpha}$ given in 1/rad.
3) **Airfoil design angle of attack and flow conditions:**
   a. Provide a summary of the C_l, C_d, C_m, and angle of attack produced at the design condition for C_l = 0.85.
   b. Provide a plot of the C_p distribution produced by the RAE 2822 airfoil at your design condition.
   c. Generate a plot showing the isocontours of the Mach number field produced by the RAE 2822 at your design condition. Take a snapshot in Paraview and include this with your submission.

Lab Instructions:

Organizing files

Create a new folder within your U:\AE 462 directory called "Lab 4". Download the "turb_RAE2822.cfg", "RAE_original_YP1_500.su2", "restart_flow_baseline.dat", and "Cook.csv" files. Place these files in your U:\AE 462\Lab 4 directory.

Converging the initial solution

We will begin the solution procedure by first converging the airfoil flow at the same conditions used in the NPARC Alliance Validation Archive, RAE 2822 Transonic Airfoil, Study #1: (https://www.grc.nasa.gov/www/wind/valid/raetaf/raetaf01/raetaf01.html). This case considers an RAE 2822 airfoil subject to freestream conditions of M = 0.729, Re = 6.5×10^6, and $\alpha = 2.31$.

Starting from the provided "restart_flow_baseline.dat" file, converge the solution. Using the surface flow data obtained generated in "surface_flow.vtu", co-plot your C_p distributions alongside the experimental data from Cook et al. You can either open the surface_flow.vtu file in Paraview and re-export as a CSV for plotting elsewhere, or co-plot directly in Paraview. For the latter case, you can develop a plot for the C_p distributions by first importing your surface_flow.vtu file and applying a "Plot Data" filter. Be sure to change the "X Axis" to the Points_X variable, instead of the default index value for the x axis. Remove all other variables such that only the C_p data is presented. Now you can import the Cook.csv file using the standard CSV reader in Paraview. Select both the PlotData and the Cook.csv items in the Pipeline Browser. Apply an "Append Attributes" filter. Click "Apply" and select the C_p plot on the right side of the screen. Change the plotting options in the Cook.csv pipeline item to properly display the experimental data. Adjust the plotting options to your satisfaction. You should see favorable agreement between your CFD solution and the experimental dataset.

Running other angles of attack

You will now run cases corresponding to other angles of attack to populate part of a lift polar. Since you already have a converged solution for $\alpha = 2.31$ deg you can use this case to warm start the solutions for other angles of attack. First, record the C_l, C_d, and C_m values produced for the baseline case at $\alpha = 2.31$ deg. [NOTE: the sign convention of C_m output from SU2 is opposite that typical of airfoil aerodynamics. Apply a negative sign to the C_m output you obtain from SU2 to correct this convention inconsistency.] In the turb_RAE2822.cfg file, change the RESTART_SOL option to YES and change the angle of attack to 1 deg. Run this case to convergence. For this and all subsequent angles of attack, record the C_l, C_d, and C_m values produced. Repeat this process to obtain a flow solution for $\alpha = 0$ deg, using your previous converged solution from $\alpha = 1$ deg to warm start this case.

Next you will run a case corresponding to $\alpha = 4$ deg. Start from the initial "restart_flow_baseline.dat" file, corresponding to $\alpha = 2.31$ deg. Increase the prescribed angle of attack to 4 deg and converge the solution. Repeat this process for $\alpha = 5$ deg. Note that these may take longer than the others to converge due to the emergence of shock-induced separation and the associated increases in flow complexity. As such, it may be best to run these cases at a time later in the week when you feel comfortable with the process and are able to dedicate a greater duration of time to run this case. $\alpha = 5$ deg is near the stall limit of the airfoil under the current freestream

conditions, which also introduces challenges in reaching a converged solution. For this case, relax the residual threshold to -5.5

CONV_RESIDUAL_MINVAL= -5.5

Return the residual threshold to -9 after completing this case.

Running based on prescribed C_l

Assume that you are configuring this airfoil section to integrate on a transonic wing geometry. Since you are intending to integrate the airfoil to produce a desired C_l, you will need to determine the geometric twist angle that matches the intended lift distribution. In this case, you are targeting a sectional $C_l = 0.85$. Rather than interpolating the coarse lift polar to determine the angle of attack for this lift condition, you can run SU2 subject to a fixed C_l condition and determine the corresponding angle of attack through the flow solution.

Open the "turb_RAE2822.cfg" file. On line 95 enable the fixed C_l mode:

FIXED_CL_MODE= YES

Change the target $C_l = 0.85$ condition on line 98. When you run SU2 it will now periodically change the angle of attack in the process of converging to the desired C_l. Once converged, record the airfoil angle of attack, C_l, C_d, and C_m produced. Also, save your flow.vtu and surface_flow.vtu files for later reporting.

You now have the CFD-generated data needed to respond to the items required for the submission.